

BUSINESS GUIDE / 2026 EDITION

AI Electricity Cost Calculator

Understanding the hidden electricity costs of AI
for UK businesses, energy and IT decision-makers.

40 TWh

UK data-centre demand by 2035*

£9,650

Enterprise annual AI energy*

46%

of DC power is compute

*Illustrative figures for guidance. See methodology and disclaimer inside.

NAVIGATION

Contents

#	Section
01	Executive Summary
02	How AI Uses Electricity
03	Understanding AI Energy Consumption
04	UK Electricity Costs Explained
05	Cost Calculator Methodology
06	Example Scenarios
07	AI vs Everyday Electricity Use
08	AI & the Future UK Grid
09	Reducing AI Energy Costs
10	Conclusions & Recommendations

How to use this guide

This report explains where AI electricity costs come from, shows the methodology behind the AcrossAI cost calculator, and gives worked UK scenarios. Use the calculator at tools.acrossai.co.uk to model your own organisation.

01 — OVERVIEW

Executive Summary

Artificial intelligence has moved from experiment to everyday business tool. Every chatbot reply, generated image and automated workflow draws electricity — first to train the underlying models, and then continuously as employees use them. For most UK organisations this cost is invisible: it sits inside cloud subscriptions and data-centre bills rather than on the company's own meter.

This guide makes that hidden consumption visible. It breaks down how AI uses electricity, explains UK business energy pricing, and sets out a transparent methodology for estimating your annual AI energy cost and carbon impact.

Why AI electricity consumption matters

- **Cost transparency:** AI energy is bundled into software pricing, making it hard for finance and energy teams to forecast true running costs.
- **Sustainability reporting:** Scope 2 and 3 emissions increasingly include cloud and AI workloads under UK SECR and ESG disclosure expectations.
- **Grid pressure:** Rapid data-centre growth is reshaping UK electricity demand and infrastructure planning.
- **Procurement decisions:** Model choice and usage patterns can change energy cost by an order of magnitude.

Key findings

~1.2 Wh per typical LLM query	17× more energy for AI images vs search	26p avg UK business price /kWh (2026*)	40 TWh projected UK DC demand by 2035*
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- A single AI text query is small (~1–4 Wh), but volume at scale is what drives cost — thousands of queries per day across a workforce add up.
- Image and especially video generation are dramatically more energy-intensive than text.
- Model selection is the single biggest lever: efficient models can deliver 10× more queries per kWh than large frontier models.
- For most SMEs, direct AI electricity cost is modest (tens to low hundreds of pounds per year); for enterprises it can reach thousands.

Future trends

- Continued double-digit growth in AI workloads and UK data-centre capacity, including designated AI Growth Zones.
- Greater pressure on grid connections, with energy availability becoming a strategic constraint on AI deployment.
- Rising importance of efficiency: smaller task-specific models, smarter caching and off-peak scheduling.

- Energy and carbon reporting moving from optional to expected in procurement and tenders.

02 — FUNDAMENTALS

How AI Uses Electricity

AI does not run on a device on your desk — it runs in large data centres packed with specialised hardware. Understanding the chain of energy use, from the building down to a single query, is the foundation for any cost estimate.

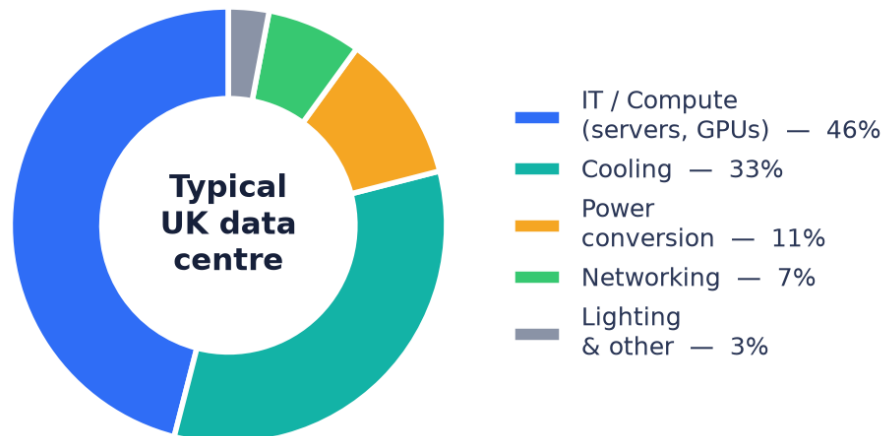


Figure 1: Approximate split of electricity use within a typical AI-capable UK data centre.

Data centres

Purpose-built facilities that host the servers running AI workloads. Their overall efficiency is measured by PUE (Power Usage Effectiveness) — the ratio of total facility energy to IT energy. Modern UK facilities target a PUE of 1.1–1.3; older sites can exceed 1.6, meaning more energy is spent on overheads.

Servers

The compute engines of AI. Training and inference rely heavily on GPUs and AI accelerators, which draw far more power than conventional CPUs — a single high-end AI server rack can demand 30–80 kW or more.

Networking equipment

Switches, routers and high-speed interconnects move enormous volumes of data between servers and to users. Networking is a smaller but constant slice of consumption.

Cooling systems

AI hardware generates intense heat. Cooling — air, liquid or immersion — is typically the second-largest energy consumer after compute, often a third of facility power.

AI model training

A one-off but very large cost: training a frontier model can consume gigawatt-hours of electricity over weeks. This is amortised across the millions of users who later rely on the model.

AI inference and everyday usage

Inference is the energy used each time the model produces an answer. Individually tiny, but because it happens on every interaction, inference is what determines your organisation's ongoing AI electricity footprint.

The bottom line

Training grabs the headlines, but for a business it is inference — your day-to-day usage — that drives recurring electricity cost. That is exactly what the AcrossAI calculator estimates.

03 — DRIVERS

Understanding AI Energy Consumption

Not all AI is equal. The energy a task consumes depends on what you are asking the model to do and how large that model is.

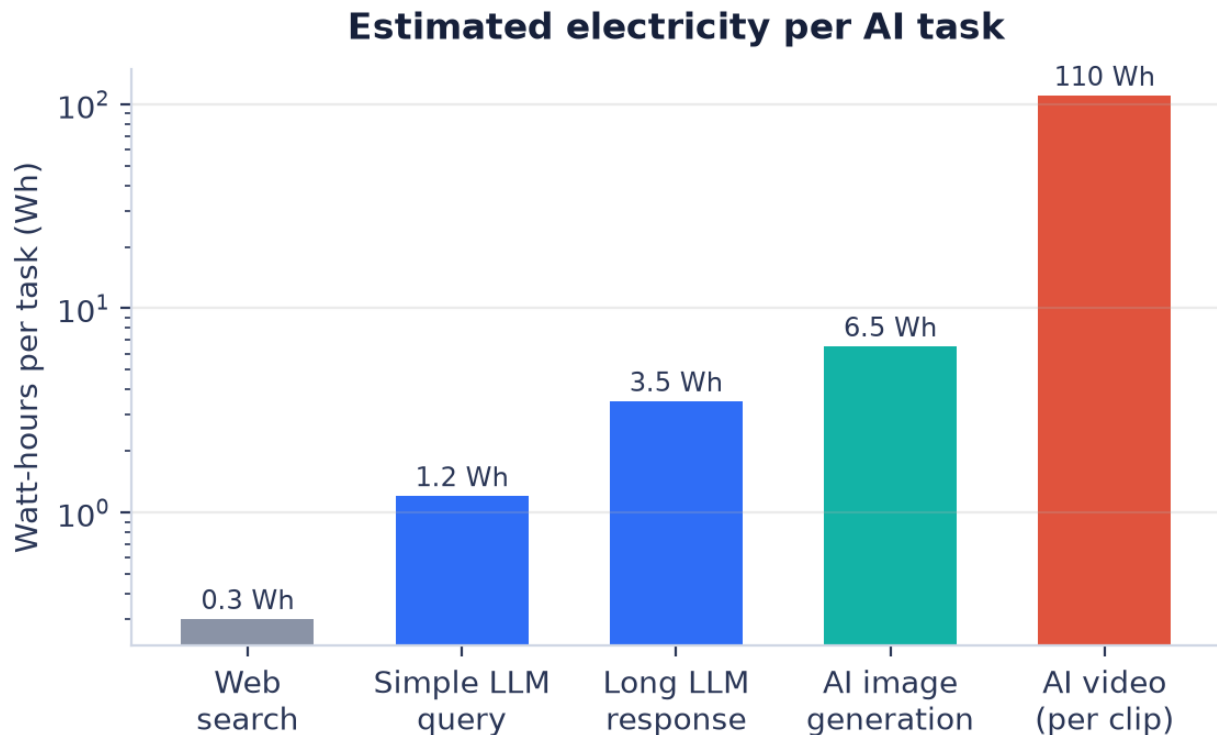


Figure 2: Indicative electricity per AI task (log scale). Video generation dwarfs text tasks.

What affects electricity use

- **Model size** — more parameters means more computation per response.
- **Output length** — longer responses run the model for more steps.
- **Task type** — text < image < video, by large margins.
- **Hardware & data-centre efficiency** — PUE and chip generation matter.
- **Usage volume** — number of users and queries per day.

Large language models (LLMs)

Text generation for chat, drafting, summarisation and coding. A short query is around 1 Wh; long, complex responses can reach several Wh. Cost is dominated by sheer volume across a workforce.

Image generation

Creating an image typically uses several times more energy than a text query (around 5–7 Wh per image), as the model runs many denoising steps.

Video generation

The most demanding common task — a single short clip can use 100+ Wh, equivalent to many hundreds of text queries.

Business AI platforms

Embedded copilots in productivity, CRM and helpdesk tools. Usage is frequent and distributed, so even small per-action energy adds up across teams.

Enterprise AI deployments

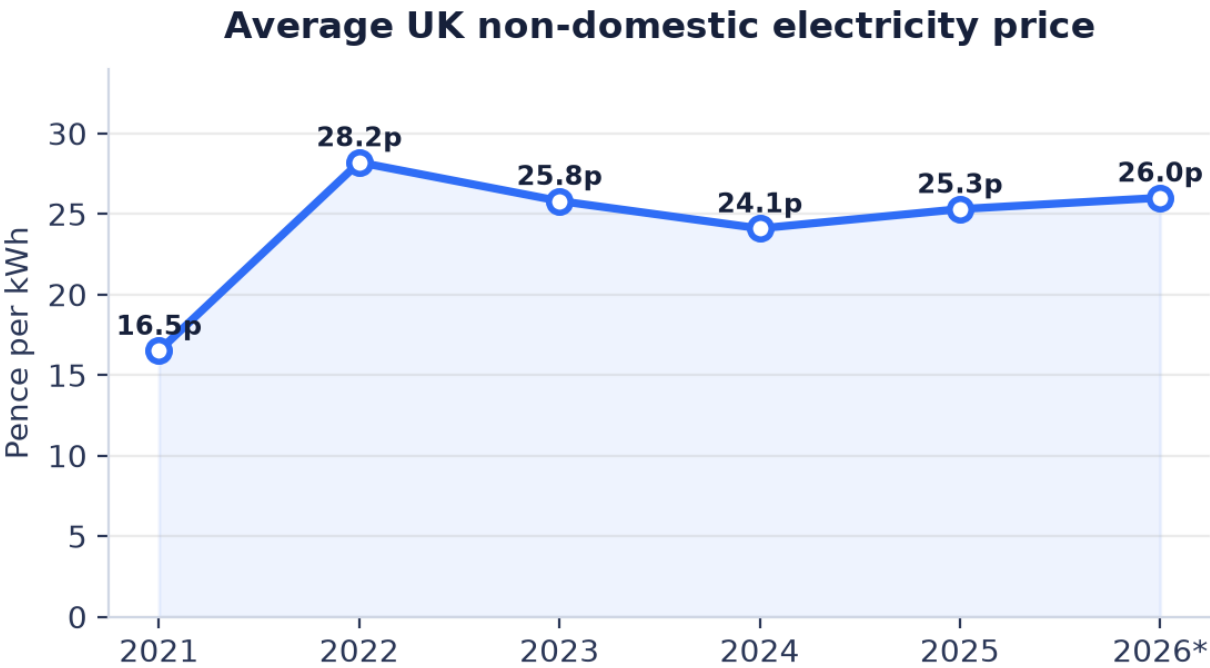
Custom models, retrieval pipelines and automation running continuously. These combine high query volumes with larger models and often dedicated infrastructure — the highest-consumption category.

AI activity	Typical energy / action	Relative intensity	Main cost driver
Simple LLM query	~1.2 Wh	Low	Volume
Long LLM response	~3.5 Wh	Low-Medium	Output length
Image generation	~6.5 Wh	Medium	Diffusion steps
Video clip	~110 Wh	Very high	Frames × steps
Enterprise pipeline	Varies (high)	High	Volume × model size

04 — PRICING

UK Electricity Costs Explained

To turn energy consumption into pounds, you need an accurate electricity price. UK business electricity pricing has several moving parts.



*2026 projected. Indicative figures for illustration.

Figure 3: Average UK non-domestic electricity unit price — indicative trend.

Business electricity pricing

Businesses pay a negotiated unit rate (pence per kWh) plus a daily standing charge. Rates vary by contract type, consumption volume, meter profile and contract length. In 2026, many SMEs pay roughly 24–28p/kWh.

Standing charges

A fixed daily fee covering grid connection and metering, independent of usage. For AI cost estimates we focus on the unit rate, since AI adds consumption rather than connection cost.

Peak vs off-peak costs

Time-of-use and half-hourly tariffs charge more during peak demand (typically late afternoon/early evening) and less overnight. Scheduling heavy or batch AI workloads off-peak can cut cost.

Regional variations

Distribution charges differ across UK regions, so the same usage can cost more in some areas than others.

Region (indicative)	Typical business unit rate*
London & South East	25–28p / kWh

Region (indicative)	Typical business unit rate*
Midlands	24-27p / kWh
North of England	24-26p / kWh
Scotland	23-26p / kWh
Wales & South West	25-28p / kWh

*Illustrative ranges for guidance only — always use your own contracted rate in the calculator.

05 — METHODOLOGY

AI Electricity Cost Calculator

The AcrossAI calculator converts your AI usage into estimated annual energy, cost and carbon. It uses a transparent, adjustable model — no black boxes.

Inputs

Input	What it captures
Employees using AI	Number of staff actively using AI tools
Daily AI usage hours	Average active AI hours per user per day
AI queries per day	Typical interactions per user per day
Type of AI activity	Text, image, video or enterprise workload
Electricity price (p/kWh)	Your contracted business unit rate

Outputs

Output	How it is derived
Annual energy (kWh)	Queries/day × Wh per query × users × working days ÷ 1000
Annual cost (£)	Annual kWh × unit rate (£/kWh)
Monthly cost (£)	Annual cost ÷ 12
Carbon impact (kgCO ₂ e)	Annual kWh × UK grid emission factor

Worked example

200 employees × 40 queries/day × 1.2 Wh = 9,600 Wh/day = 9.6 kWh/day. Over ~250 working days = 2,400 kWh/year. At 26p/kWh that is roughly £624/year, plus a carbon estimate using the UK grid factor (~0.21 kgCO₂e per kWh).

The model is deliberately conservative and adjustable. Energy-per-task values are estimates that can be refined as your usage mix changes. The UK grid emission factor falls each year as generation decarbonises, so carbon estimates improve over time.

06 — WORKED EXAMPLES

Example Scenarios

Three illustrative UK scenarios show how usage, scale and activity type drive very different outcomes.

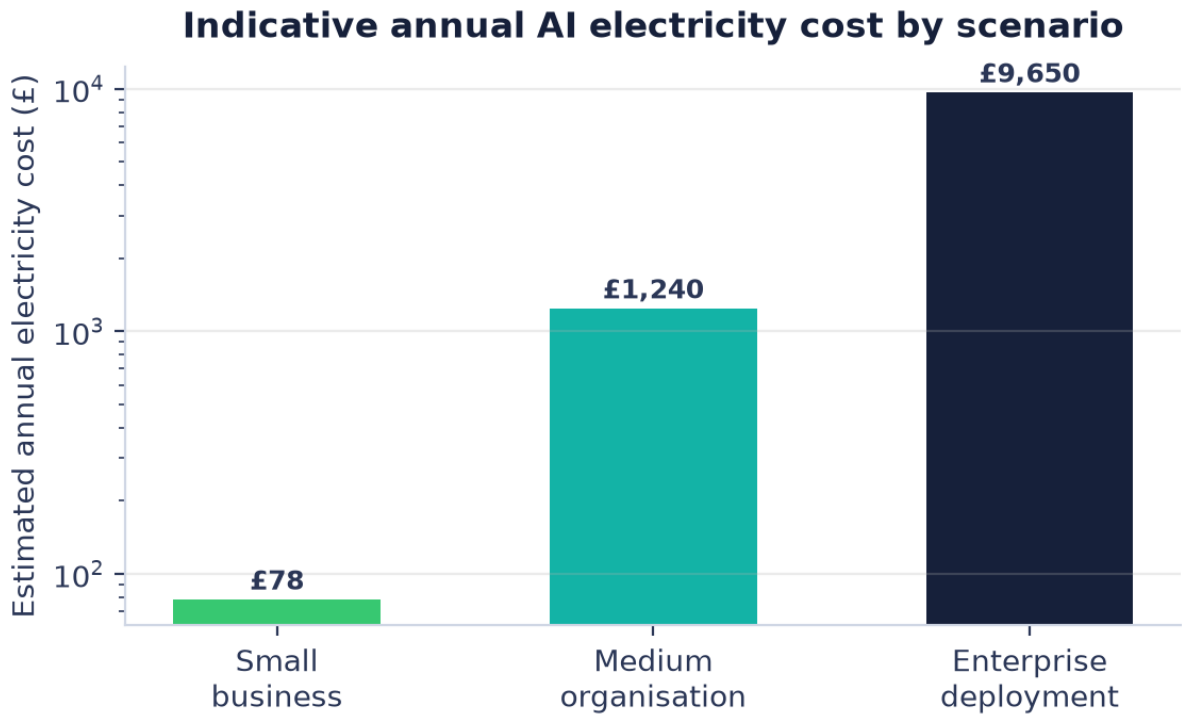


Figure 4: Indicative annual electricity cost by scenario (log scale).

Scenario 1 — Small business using ChatGPT

A 10-person consultancy where staff use an AI assistant for drafting and research, ~30 queries each per day. Light, text-only usage.

Scenario 2 — Medium organisation, AI customer service

A 150-seat company running an AI-enabled support desk handling thousands of conversations daily, with some image and document processing.

Scenario 3 — Enterprise-scale AI deployment

A large enterprise with custom models, automation pipelines and high query volumes running continuously across departments.

Metric	Small business	Medium org	Enterprise
Users	10	150	2,000+
Queries / day (total)	300	12,000	250,000+
Primary activity	Text	Text + images	Mixed + pipelines
Annual energy	~300 kWh	~4,800 kWh	~37,000 kWh
Annual cost*	~£78	~£1,240	~£9,650

Metric	Small business	Medium org	Enterprise
Monthly cost*	~£6.50	~£103	~£804

*Illustrative, at 26p/kWh. Actual figures depend on model choice, usage mix and your contracted rate.

07 — CONTEXT

AI vs Everyday Electricity Use

Putting AI energy in everyday terms helps stakeholders grasp its real scale. The surprise for many is how small individual AI tasks are — and how the cost lives in volume, not single actions.

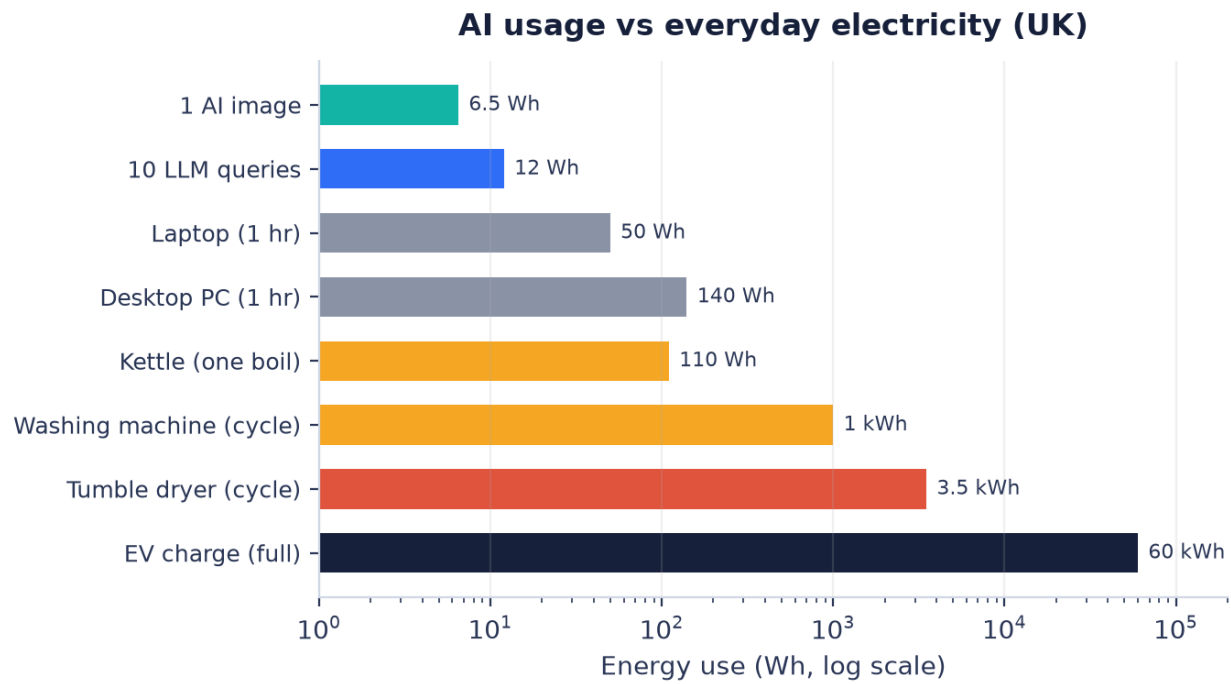


Figure 5: AI tasks compared with common UK household and office appliances (log scale).

Activity	Energy use	Equivalent
10 LLM queries	~12 Wh	About 1 minute of a kettle
1 AI image	~6.5 Wh	A few minutes of a laptop
Boiling a kettle	~110 Wh	≈ 90 LLM queries
Washing machine cycle	~1 kWh	≈ 830 LLM queries
Tumble dryer cycle	~3.5 kWh	≈ 2,900 LLM queries
Full EV charge	~60 kWh	≈ 50,000 LLM queries

Reframing the conversation

One AI query costs a tiny fraction of a penny. The real question for businesses is not 'is one query expensive?' but 'what does our total usage, at scale, cost across a year?'

08 — OUTLOOK

AI & the Future UK Electricity Grid

AI's biggest energy story is not the individual query — it is aggregate demand. Data centres are becoming one of the fastest-growing loads on the UK grid.

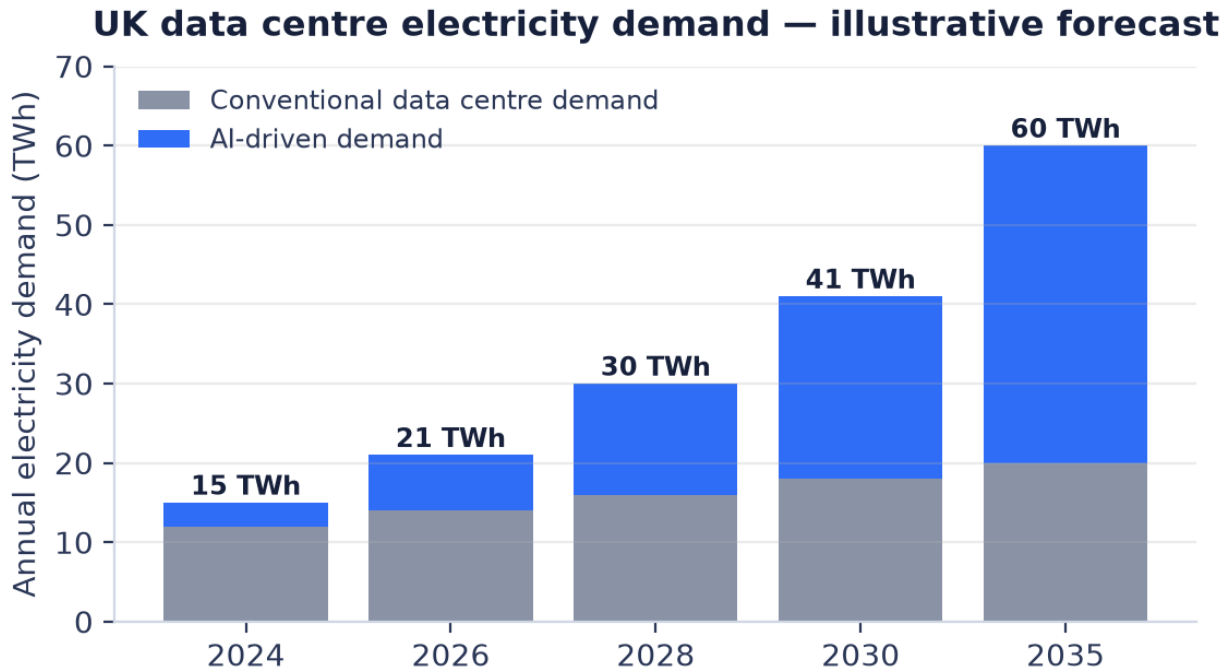


Figure 6: Illustrative UK data-centre electricity demand, split by AI-driven growth.

Future demand forecasts

Data centres currently account for a small but rising share of UK electricity. AI is the main growth driver, with demand potentially several times higher by the mid-2030s.

Data centre growth

Operators are expanding capacity rapidly, concentrated around London, the South East and emerging regional hubs with available power and connectivity.

AI Growth Zones

UK Government policy is designating AI Growth Zones to accelerate compute capacity, pairing planning support with energy infrastructure to attract investment.

Grid infrastructure challenges

- Connection queues: securing grid capacity is now a key constraint on new data centres.
- Local network reinforcement needed to support concentrated high-density loads.
- Balancing AI demand growth with the UK's net-zero and decarbonisation targets.
- Rising interest in on-site generation, storage and flexible/off-peak operation.

09 — ACTION

Reducing AI Energy Costs

The good news: AI energy cost is highly controllable. Most savings come from smarter usage and model choice rather than doing less AI.

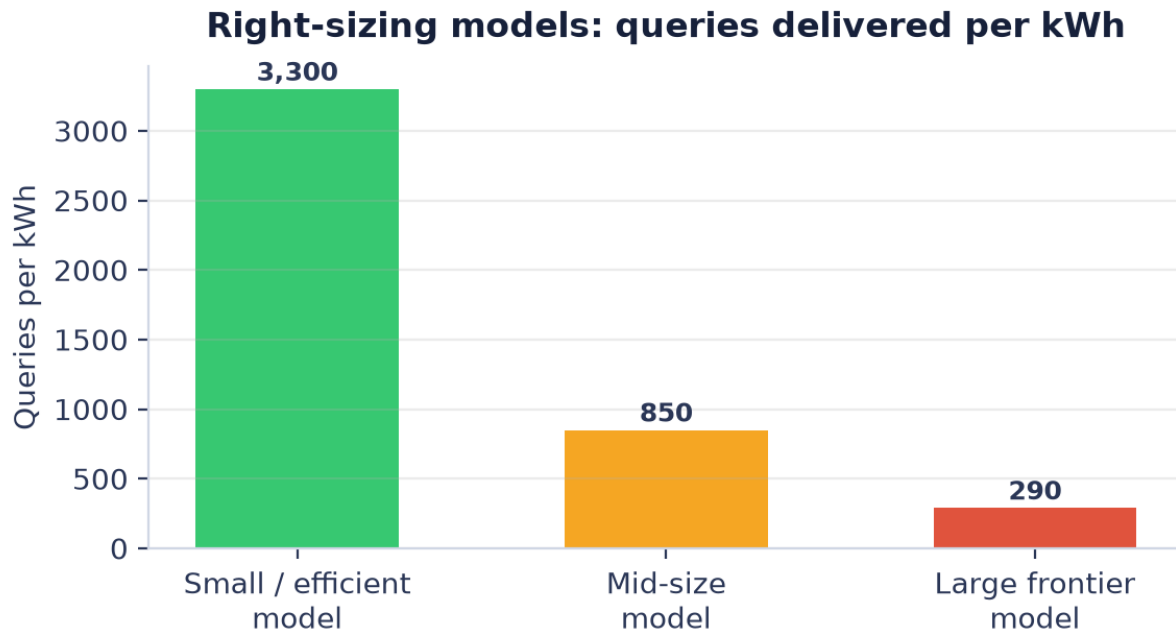


Figure 7: Right-sizing models can deliver many more queries per kWh.

Efficient AI use

- Reuse and cache common answers instead of regenerating them.
- Write tighter prompts to avoid unnecessarily long outputs.
- Batch non-urgent and bulk jobs to run off-peak.

Model selection

- Match the model to the task — use small, efficient models for routine work.
- Reserve large frontier models for genuinely complex problems.
- Favour providers and regions powered by low-carbon electricity.

Infrastructure optimisation

- Choose data centres with low PUE and high renewable supply.
- Use modern, efficient AI accelerators and right-sized capacity.
- Consolidate workloads to improve hardware utilisation.

Energy management strategies

- Track AI usage and energy as part of sustainability reporting.
- Set internal guidelines on heavy tasks like video generation.
- Review contracted electricity rates and time-of-use tariffs.
- Use the AcrossAI calculator to baseline and monitor over time.

10 — SUMMARY

Conclusions & Recommendations

AI is now a permanent line item in business operations — and it carries a real, if often hidden, electricity cost. Organisations that measure and manage it gain both financial control and a credible sustainability story.

Key takeaways

- Individual AI tasks are cheap; cost and carbon scale with total usage volume.
- Activity type matters enormously — text is light, video is heavy.
- Model selection is the single biggest cost-and-carbon lever available.
- UK data-centre demand is rising fast, making efficiency strategically important.
- Transparent measurement turns hidden cost into a manageable metric.

Business recommendations

Priority	Recommended action
1	Baseline your AI energy and cost using the AcrossAI calculator.
2	Set model-selection guidelines to default to efficient models.
3	Schedule heavy/batch workloads off-peak where possible.
4	Include AI energy in Scope 2/3 sustainability reporting.
5	Review electricity contracts and regional/time-of-use options.

Estimate your own AI electricity cost

Model your organisation's AI energy, cost and carbon with the free interactive tools at tools.acrossai.co.uk — and explore the full suite of AI energy and infrastructure calculators from AcrossAI.

Disclaimer

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